UAT: Exploring Latent Uncertainty for Semi-Supervised Object Detection in Remote-Sensing Imagery

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***Abstract*— Object detectors for remote sensing imagery (RSI) are plagued by data-driven training approaches, underperforming when confronted with inadequate object information. Thus, semi-supervised object detection (SSOD) is introduced to alleviate high data dependence. However, it is unfeasible to directly apply existing SSOD methods, considering the classification and regression uncertainties resulting from unique characteristics of RSI: 1) interclass uncertainty: objects in remote sensing scenes are usually densely arranged, leading to severe feature fusion between different categories; 2) intraclass uncertainty: for same-category objects, the complicated contexts induced by various scenarios increase intra-class feature diversity, posing a challenge to clustering; and 3) regression uncertainty: the inherent scale variation in RSI gives rise to inconsistent convergence rate, hindering the regression progress from effective converging. Therefore, we propose a teacher–student model (TSM)-based SSOD method aiming at remote sensing scenes, termed uncertainty-aware teacher (UAT), to quantify and promote the detector's confidence, which is composed of entropy-based uncertain-pair softening (EUS) policy, multi-Gaussian sample fitting (MSF) module and scale- adaptive adjunct loss (SAL). Specifically, EUS identifies the uncertain category pairs with class-wise information entropy, MSF remodels the intraclass distribution and sets the optimal threshold dynamically for pseudo-labels, and SAL adaptively rescales bboxes based on the original size to accelerate the regression convergence. We have proven the efficiency of our method through extensive experiments on two public remote sensing datasets, DOTA and DIOR.**

***Index Terms*— Remote sensing imagery (RSI), semi-supervised object detection (SSOD), teacher–student model (TSM).**

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I. Introduction

REMOTE sensing imagery (RSI) are photographs captured from an aerial perspective, broadly encompassing diverse scenes on the Earth's surface. With the capacity to classify and localize simultaneously, object detection [1] is extensively applied to RSI [2], [3], serving as a fundamental technology for several essential applications, e.g., Earth observation and space exploration [4], [5], [6]. Consequently, the research on remote sensing object detection (RSOD) is flourishing with increasing attention [7], [8], [9].

Practically, most RSOD methods require a large amount of labeled data to improve the model performance [10], [11], [12], but the expensive availability of manual annotations hinders the development of these labeled data-driven detectors. Hence, semi-supervised learning [13] is introduced to assist the models in delving unlabeled data that carry voluminous unexplored information in remote sensing tasks [14], [15], [16]. The teacher–student model (TSM) is commonly employed as semi-supervised object detection (SSOD) methods [17], [18], as illustrated in Fig. 1, where an unsupervised branch is utilized to explore the potential knowledge of unlabeled data. Based on the dual-branch structure, pseudo-labels are generated by the teacher to supervise the student, and the teacher model's parameters are iteratively updated from the student by exponential moving average (EMA) [19]. In recent years, several researchers have demonstrated that SSOD is effective and reliable in achieving excellent performance, greatly reducing the annotation cost.

However, the huge misalignment between natural images and RSI resulting from the inherent data properties hinders general SSOD methods from directly applying to RSI, resulting in doubtful prediction confidence. This unreliable confidence, defined as model uncertainty, leads to an undesirable degree of models' trustworthiness in SSOD. As shown in Fig. 1, the latent model uncertainty arising from data-specific traits mainly exists in the teacher model, especially influencing the backpropagation process. To thoroughly analyze the model uncertainty, we first decompose it into classification and regression uncertainties to adapt to object detection tasks, and then introduce the specific characteristics of RSI and their corresponding impacts under semi-supervised conditions as follows.

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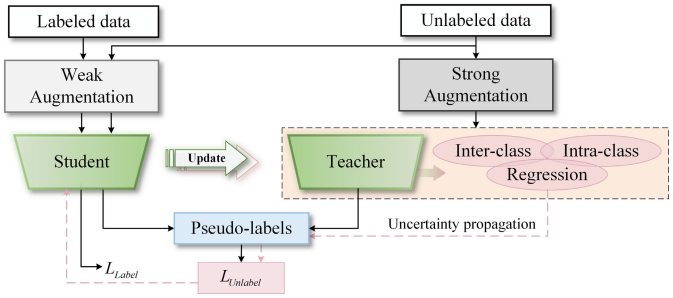


Fig. 1. Framework of TSM. (Left) Pipeline of the supervised branch, while (right) unsupervised one. In the unsupervised branch, the pink ellipses and dashed arrows represent the latent uncertainties in the teacher model and its impact during backpropagation, respectively.

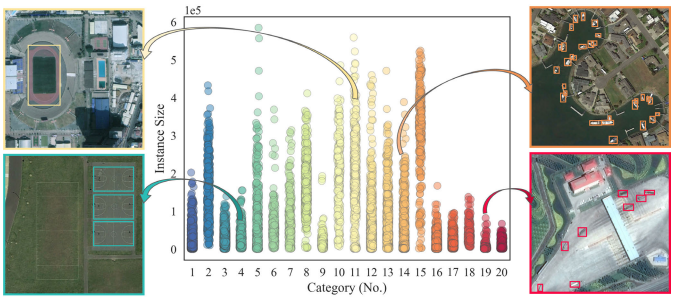
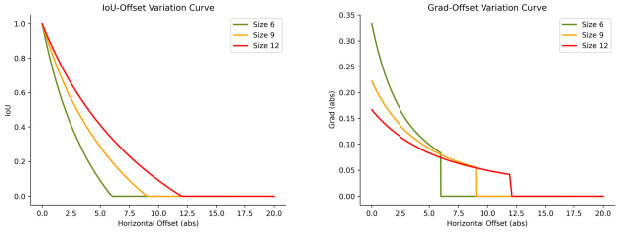


Fig. 2. Medium figure illustrates each category's instance size distribution on the DIOR training set, and the corresponding category name can be found in Table II. (Left and right) parts give some examples of objects with different scales.



(a) (b)

Fig. 3. IoU/gradient-offset variation curves of bboxes with different scales. The green, yellow, and red curves refer to bboxes with size of 6, 9, and 12, respectively. "Horizontal offset" means the offset between the bbox and the GT in the horizontal direction, which represents different relative location relationships. In (a) and (b), vertical axes represent IoU and gradient, respectively.

A. Interclass Uncertainty

In RSI, most instances are densely distributed, exerting overlapping between objects, especially for horizontal bounding box (HBB) RSOD. Consequently, samples from distinct classes occupy similar feature presentations in the feature space, mathematically resulting in decision boundaries that are either shifted or exhibit instability (e.g., jitter or oscillation). This interclass uncertainty decreases the model's category discrimination ability, especially for the teacher to generate controversial pseudo-bboxes, inducing unreliable supervision for the student.

B. Intraclass Uncertainty

In contrast to natural images, RSI exhibits a wide range of backgrounds that dominate the majority of an image (e.g., forests, roads, and water). Interfered by complex surrounding contexts, it is possible for the detectors to interpret same-category objects as features with discrepancies. This intraclass uncertainty simultaneously brings noise to the teacher and student, leading to conflicting interpretations.

C. Regression Uncertainty

As illustrated in Fig. 2, RSI exhibit a diverse range of object scales attributed to the unique bird's-eye view. Thus, a convergence inconsistency of multiscale objects exists in the regression progress, causing misaligned regression rates. Fig. 3 analyzes how the scale variations of bboxes affect the degree of change in intersection-over-union (IoU) and gradient under the same offset. In Fig. 3(a) and (b), the similar trends of IoU and gradient change suggest the model's regression sensitivity, which is correspondingly influenced by drastic fluctuations in bbox scales.

On account of the RSI challenges above, several researchers have investigated the tremendous potential in unlabeled data for RSOD [20], [21], [22]. Semi-supervised methods effectively reduce annotation costs while maximizing data utilization, verifying that implementing semi-supervised RSOD merits further research. Thus, we propose a TSM-based SSOD method, uncertainty-aware teacher (UAT), which explores latent uncertainty in inter- and intraclass feature interpretation and scale-aware regression convergence. To the best of our knowledge, UAT is a pioneer in evaluating the uncertainty of SSOD models in separate tasks for RSI, quantifying the detector's reliability, and promoting it.

For classification uncertainty, we resolve it mainly in the teacher to ensure feature quality and rectify pseudo-bboxes guidance for the student, from interclass and intraclass perspectives, respectively. Specifically, to alleviate interclass feature confusion stemming from dense arrangement in RSI, we design an entropy-based uncertain-pair softening (EUS) policy. EUS quantifies the clustering complication assisted by information entropy and then provides interpretable softened labels of uncertain category pairs to fine-tune the model's attention. Considering that scenario complexity poses challenges for precise same-category identification, a multi-Gaussian sample fitting (MSF) module is devised to depict intraclass diversity by excavating ambiguous marginal features for each category. MSF dynamically customizes the soundest threshold for the student to filter out higher quality pseudo-bboxes.

For regression uncertainty, a scale-adaptive adjunct loss (SAL) is introduced to adjust the scale-relevant discrepant regression in the student, optimizing the model's convergence robustness to scale variety. Particularly, SAL designs specific adjunct bboxes for high/low IoU samples with an adaptive scaling ratio. Further integrated into the original IoU loss, it empowers the detector with expedited convergence depending on the actual regression state. Our main contributions are summarized as follows.

1) We propose a novel SSOD method UAT for RSI which evaluates the model uncertainty in classification and

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regression tasks, notably improving the feature discrimination ability of the model and effectively accelerating the regression process. Our work has been proven to exhibit SOTA performance.

2) EUS is designed to improve the feature reliability in the teacher, discriminating the instances with dense arrangement, and balancing the model's attention to the confusable categories.
3) To improve the detector's scenario adaptability, MSF is devised to explore the intraclass diversity and reasonably offer higher quality pseudo-bboxes for the student model.
4) Based on the conflicting regression convergence caused by scale variation, SAL is introduced to enhance the scale sensitivity of the student in the regression procedure.

The rest of this article is organized as follows. Section II reviews previous RSOD works and the utilization of unlabeled data in RSI. In Section III, we furnish a detailed account of UAT, unpacking each component thoroughly. Section IV demonstrates and analyzes the experimental results in detail. The conclusion of our work is presented in Section V.

II. Related Work

A. Remote Sensing Object Detection

Object detection is considered a fundamental and essential task in computer vision (CV). Mainstream detectors are commonly grouped into two-stage-based and one-stage-based. Confronted with the inherent characteristics of RSI, e.g., complex background, dense distribution, scale variation, and arbitrary orientation, tailored RSOD methods have been widely investigated and exhibit outstanding performance [23].

For instance, Zhang et al. [24] propose PIIDet, a novel one-stage anchor-free detector, to generate specific-category instances and accurately generate OBBs in complex backgrounds. Gao et al. [25] propose a scale-aware detector GLSANet that designs an attention pyramid to enhance the representation of small objects and suppress the useless background noise. To address the issue of object density, CoF-Net [26] refines the course features into fine features by discriminating their semantic relations. For oriented object detection (OOD), Zhang et al. [27] propose FANet, innovatively fusing cross-level features based on feature pyramid network (FPN) Lin et al. [28]. UniconDet [29] is a unified object contour detector which achieves multigrained object detection, utilizing Fourier transformation in the frequency domain.

With the evolution of transformer in object detection [30], many researchers have devised specific transformer-based methods for RSOD, exploiting its ability to capture long-range information. For example, Wang et al. [31] integrate transformer blocks into the convolutional neural network (CNN), leveraging the strengths of both and enhancing multiscale feature aggregation. FPNFormer [32] effectively improves the expressiveness of the model with a bidirectional fusion FPN, dealing with the object orientation problem. However, these methods necessitate considerable annotated data for training, severely hindering the conduct of RSOD research.

Several strategies have been employed to mine latent information from limited data to overcome the data gap in large models. Gao et al. [33] introduce few-shot learning to RSI, handling the scale variation with channel multiplication utilizing FPN and Transformer. Antonelli et al. [34] address the misidentification due to partial annotation, promoting the advancement of few-shot object detection (FSOD) in RSI.

B. Unlabeled Data Exploration in RSOD

The exploration of unlabeled data in RSOD has undergone a remarkable journey, evolving from initial challenges to sophisticated techniques and methodologies. With the purpose of reducing annotation expenses, self-supervised learning enables models to generate supervision signals through designing pretext tasks, such as SSAFN [35], which tackles the struggle of background complexity by constraining the foreground enhancement through self-supervised tasks. To eliminate the scene- and object-level inconsistency between pretraining and fine-tuning, Zhang et al. [36] propose an object-centric strategy based on masked image modeling. Additionally, weakly supervised object detection (WSOD) methods are introduced to improve data utilization efficiency, alleviating the tremendous demand for fine-labeled data. Qian et al. [37] mine high-quality instances with a novel metric, measuring the proposal quality to describe the object covering completeness. Erasing the most distinct region and balancing the instance IoU, Xie et al. [38] capture the integral object content and mine more low-quality instances.

Another exploration direction is SSOD, where vast amounts of unlabeled data are fully leveraged to unleash the immense potential, compensating for labeled data and learning generalizable representations of image content. PCT [20] strengthens the sensitivity of multi-scale objects and enhances the quality of pseudo-labels, taking full advantage of unlabeled data and serving as a novel SSOD method in RSI based on TSM. Nevertheless, the semi-supervised methods for RSOD remain ripe for further exploration. It is essential to undertake a deeper investigation of SSOD in RSI.

III. Proposed Method

Existing semi-supervised methods are struggling with the high complexity of RSI due to clustered arrangement and severe scale variation. Motivated by consistent-teacher [18], we propose a TSM-based approach, namely UAT, which measures the uncertainties of SSOD models from two aspects: classification and regression, boosting the trustworthiness of the model predictions to the utmost.

We divide this section into four parts. First, we clarify the network architecture, and the next three parts elaborate on the core components, i.e., EUS, MSF, and SAL. The whole framework is illustrated in Fig. 4.

A. UAT Architecture

Based on the vanilla TSM framework, we propose an uncertainty-aware SSOD method UAT to comprehensively evaluate the uncertainties in remote sensing tasks, assessing

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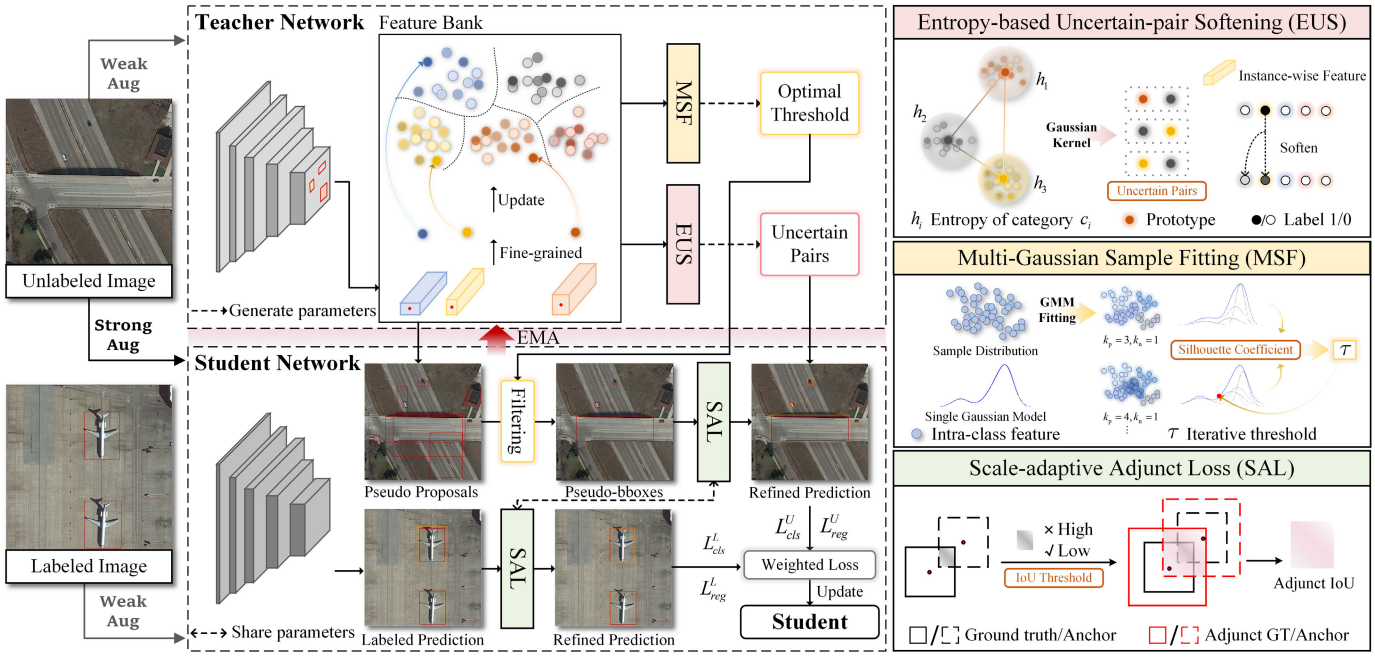


Fig. 4. Overview of UAT. Our proposed framework comprises three core modules, namely EUS, MSF, and SAL. EUS dynamically measures class-wise information entropy and prototypes and softens the uncertain-pair labels to reduce interclass uncertainty. MSF, guided by multi-Gaussian models, fits the positive sample distribution and iterates an optimal threshold to mitigate intraclass uncertainty. SAL introduces adjunct bboxes depending on the explicit relationship between IoU and the object scale to alleviate regression uncertainty.

and elevating the model performance. As shown in Fig. 4, the overall architecture comprises two networks, the teacher and the student. The labeled image $\mathbf{I}_l$ is processed through a common supervised RSOD method and the teacher model is updated by EMA from the student. The input unlabeled image $\mathbf{I}_u$ is first weakly augmented and sent to the teacher for detailed feature extraction. The refined proposal features are then fine-grained by central pixel extraction and updated into the feature bank, making provisions for generating pseudo-labels as the supervision of predictions of the strongly augmented $\mathbf{I}_u$ in the student.

The whole SSOD training procedure is shown as Algorithm 1, where we simplified the data augmentation and in-batch aggregation. Based on the generic training strategy, we integrated our uncertainty-aware modules to optimize TSM for RSI. Given the labeled dataset $\mathbf{D}_l = \{\mathbf{x}_i^l, \mathbf{y}_i^l\}^{N_l}$ and the unlabeled dataset $\mathbf{D}_u = \{\mathbf{x}_j^u\}^{N_u}$, where $\mathbf{y}_i^l = \{y_n = (c_n, b_n^{\text{gt}})\}_{n=1}^{N_i}$ is the ground truth (GT) of the ith image with N_i instances, similarly, we denote $\hat{\mathbf{y}}_j^u$ as the pseudo-labels annotated by the teacher. The original classification loss and regression loss are set as focal loss [39] and CIoU loss [40], respectively. Refined by EUS and SAL, the losses take the uncertain category pairs $\{c_{\text{ua}} = < c_i, c_j >\}$ and adjunct bboxes $\hat{b}$ into account to estimate the task-oriented uncertainties. Thus, the losses of UAT are described as follows:

$$L_{\text{label}} = \frac{1}{N_l}\sum_{i}^{N_l}\left[L_{\text{cls}}^{l}\left(\mathbf{x}_i^l, \mathbf{y}_i^l, c_{\text{ua}}\right) + L_{\text{reg}}^{l}\left(\mathbf{x}_i^l, \mathbf{y}_i^l, \hat{b}_i\right)\right] \quad (1)$$

$$L_{\text{unlabel}} = \frac{1}{N_u}\sum_{j}^{N_u}\left[L_{\text{cls}}^{u}\left(\mathbf{x}_j^u, \hat{\mathbf{y}}_j^u, c_{\text{ua}}\right) + L_{\text{reg}}^{u}\left(\mathbf{x}_j^u, \hat{\mathbf{y}}_j^u, \hat{b}_i\right)\right] \quad (2)$$

$$L_{\text{total}} = L_{\text{label}} + \lambda_u L_{\text{unlabel}} \quad (3)$$

Algorithm 1 SSOD Training Strategy

Inputs:
$\mathbf{D}_l$: Labeled dataset
$\mathbf{D}_u$: Unlabeled dataset
λ_u: Weight for the unsupervised branch
K : Number of iterations
f_{θ_t} : Teacher model
f_{θ_s} : Student model

Outputs:
θ_t^* : Updated teacher model parameters
θ_s^* : Updated student model parameters

Procedure:
for iteration = 1 to K **do**
$\{(\mathbf{x}_i^l, \mathbf{y}_i^l)\} \leftarrow \text{RandomSample}(\mathbf{D}_l)$
$\{\mathbf{x}_i^u\} \leftarrow \text{RandomSample}(\mathbf{D}_u)$
$p^{pseudo} \leftarrow f_{\theta_t}(\mathbf{x}_i^u)$
$p^l \leftarrow f_{\theta_s}((\mathbf{x}_i^l, \mathbf{y}_i^l))$
$p^u \leftarrow f_{\theta_s}(\mathbf{x}_i^u)$
$\theta_s^* \leftarrow L_{label}(p^l, \mathbf{y}^l) + \lambda_u L_{unlabel}(p^u, p^{pseudo})$
$\theta_t^* \leftarrow EMA(\theta_s^*)$
end for
return θ_s^*, θ_t^*

where the specific definition of $\hat{b}$ is clarified in Section III-D and λ_u is manually set as 0.2 to weigh the unsupervised branch.

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B. Entropy-Based Uncertain-Pair Softening

In RSI, objects are densely distributed, which means the boundaries of HBBs may overlap with each other, leading to severe feature confusion and hindering the detector from clustering distinct categories. Thus, the interclass similarity induces vogue borders, intensifying the classification uncertainty. To mitigate the blurry issue, we devise a simple but effective approach motivated by information entropy and prototype learning [41], excavating the indeterminate chaotic and mutually confusable categories, namely uncertain category pairs c_{ua}. Subsequently, we soften the corresponding labels pairwise in the classification head only if the current instance is recognized as one of c_{ua}, therefore assisting the model in discovering more explicit interclass relationships and improving feature discrimination capability.

A feature bank $\mathbf{F}_c = \{\mathbf{f}_i, s_i\}^{N_f}$ is built to store continually updated instance-wise features for each category c, where s_i is the confidence score and N_f is set at 100 to save memory. In each iteration, the proposal feature $\hat{\mathbf{f}}_i$ is first filtered by the confidence threshold τ_{eus} to reduce the negative impact of low-credibility features, and then fine-grained through extracting the feature vector with spatial size of 1×1 at the center of $\hat{\mathbf{f}}_i$, which is named as central pixel extraction, as $\mathbf{f}_i$ to ensure feature reliability. Subsequently, the category-wise information entropy h_c is calculated to quantify the degree of chaos and the average entropy h_m sifts out the uncertain categories $\{c'\}^{N_{c'}}$, where $h_{c'} > h_m$ and $N_{c'}$ refers to the number of uncertain categories. h_c and h_m are calculated as follows:

$$h_c = -\sum_{\mathbf{f}_i \in \mathbf{F}_c} s_i \log s_i \tag{4}$$

$$h_m = \frac{1}{N_c} \sum_{c}^{N_c} h_c \tag{5}$$

where N_c refers to the number of total categories and follows $N_{c'} < N_c$.

Then, we compute the class-wise prototypes to construct the prototype distance matrix (PDM) $\mathbf{M}_D \in \mathbb{R}^{N_{c'} \times N_{c'}}$ to quantify the interclass separation of $N_{c'}$ categories, in which each element d_{ij} signifies the distance between class-wise prototypes p_i and p_j that obtained by averaging all the features in the corresponding feature bank, i.e., the highly abstract representative vectors of category c_i and c_j

$$\mathbf{M}_D = \begin{bmatrix} 0 & K(p_1, p_2) & \cdots & K\left(p_1, p_{N_{c'}}\right) \\ K(p_2, p_1) & 0 & \cdots & K\left(p_2, p_{N_{c'}}\right) \\ \vdots & \vdots & \ddots & \vdots \\ K\left(p_{N_{c'}}, p_1\right) & K\left(p_{N_{c'}}, p_2\right) & \cdots & 0 \end{bmatrix} \tag{6}$$

where $K(\cdot)$ refers to Gaussian kernel function. Besides, $K(p_i, p_i)$ is set as 0 to avoid redundancy calculation. Finally, we construct uncertain pairs by combining each c_i with c_j which meets the following condition:

$$< c_i, c_j > = \arg\min_{j, j \neq i} \left\{ K\left(p_i, p_j\right) \right\}. \tag{7}$$

Since $\{c_{\text{ua}} = < c_i, c_j >\}$ represents the categories that the model has cognitive defects in understanding at different training stages, we combine label smoothing [42] strategy to decrease the model's confidence and directly face the uncertainty. For the current detected instance sample x that belongs to category c_i with a GT label y, we denote the GT label distribution $Q(c_i|x) = \delta_{i,y}$, where $\delta_{i,y}$ prefers to Dirac delta. Specifically, when c_i belongs to c_{ua}, we introduce η to soften the uncertain labels in the classification head. For example, Q is revised as $\tilde{Q}$

$$\tilde{Q}(c_i|x) = \eta \delta_{i,y} + (1 - \eta)\delta_{j,y}. \tag{8}$$

Considering the potential negative effects of EUS, we have limited its lifespan during the training phase to minimize latent confusion, particularly during the model's converging stage.

C. Multi-Gaussian Sample Fitting

Considering the existence of complex backgrounds in RSI, objects with the same category may exhibit distinct context information under specific scenarios. These regions are mostly irregular owing to environment components, imaging conditions, and weather patterns, severely compromising the model's descriptive integrity, i.e., the model has difficulty distinguishing features on category boundaries. Hence, a multi-Gaussian-based approach MSF is proposed to perfect the model's dynamic perception of class-surrounding features in different stages, mitigating the model's erroneous classification decisions.

For a specific category c_i, MSF hypothesizes that all unlabeled instances with confidence scores s^i are Gaussian mixture-distributed in negative $\mathcal{N}_n^i$ and positive $\mathcal{N}_p^i$ modalities. Furthermore, we decompose $\mathcal{N}_p^i$ with multi-Gaussian models to simulate the complicated inner subcategory distribution, and the positive Gaussian number K^i depends on the actual distribution pattern

$$\mathcal{D}\left(s^i\right) = \alpha^i \mathcal{N}_n^i\left(s^i \middle| \mu_n^i, \left(\sigma_n^i\right)^2\right) + \sum_{1}^{K^i} \beta^i \mathcal{N}_p^i\left(s^i \middle| \mu_p^i, \left(\sigma_p^i\right)^2\right) \tag{9}$$

where $\mathcal{N}(\mu, (\sigma)^2)$ refers to a Gaussian distribution, α_n^i and β_p^i denote the weights that reflect the relative importance of each Gaussian, following the rule of $\alpha^i + \sum\{\beta^i\}^{K^i} = 1$. Notably, to mitigate the unnecessary influence of interference information from complex backgrounds, a single Gaussian model is employed to model the negative samples in MSF.

We hypothesize that category c_i can be split into K^i subcategories $\{c_k\}^{K^i}$, where K^i is defined by (14). During the iteration, we first determine the cluster number K' to obtain $\{c_k\}$ by K-means, where $k = 1, 2, \ldots, K'$. Then, we infer the Silhouette Coefficient $\text{SC}^{K'}$ that combines the clustering cohesion $\mathbf{H}$ and separation $\mathbf{P}$ to evaluate the model's descriptive ability of intraclass diversity. For feature $x_i \in c_k$, $\mathbf{H}(x_i)$ and $\mathbf{P}(x_i)$ are written as

$$\mathbf{H}(x_i) = \Phi\left(\Sigma_{x_j \in c_k, x_i \neq x_j} \rho\left(x_i, x_j\right)\right) \tag{10}$$

$$\mathbf{P}(x_i) = \min \Phi\left(\Sigma_{x_j \in c_k', c_k' \neq c_k} \rho\left(x_i, x_j\right)\right) \tag{11}$$

where $\Phi(\cdot)$ and $\rho(\cdot)$ refer to the average function and Euclidean distance, respectively, and $c_k' \subset c_i$. Thus, the $\text{SC}^{K'}$

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can be inferred by adhering to

$$\mathrm{SC}^{K'} = \max\Phi\left(\Sigma s_{\mathrm{sil}}^{K'}(x_i)\right) \tag{12}$$

$$s_{\mathrm{sil}}^{K'}(x_i) = \begin{cases} 1 - \dfrac{\mathbf{H}(x_i)}{\mathbf{P}(x_i)}, & \mathbf{H}(x_i) < \mathbf{P}(x_i) \\ 0, & \mathbf{H}(x_i) = \mathbf{P}(x_i) \\ \dfrac{\mathbf{P}(x_i)}{\mathbf{H}(x_i)} - 1, & \mathbf{H}(x_i) > \mathbf{P}(x_i) \end{cases} \tag{13}$$

where $\mathrm{SC}^{K'} \in [-1, 1]$. The larger $\mathrm{SC}^{K'}$ is, the more reasonable the current distribution is, representing the model's fine-grained discriminative ability. The final K is defined as

$$K^i = \arg\max_{K'} \mathrm{SC}^{K'}. \tag{14}$$

At last, the expectation–maximization (EM) algorithm is adopted to fit the final positive posterior $\mathcal{D}_p(x|s^i, \mu_p^i, (\sigma_p^i)^2)$, and the class-specific adaptive threshold τ^i to filter the pseudo targets for the student

$$\tau^i = \arg\max_{s^i} \mathcal{D}_p\left(x \middle| s^i, \mu_p^i, \left(\sigma_p^i\right)^2\right). \tag{15}$$

D. Scale-Adaptive Adjunct Loss

IoU-based loss functions are widely applied to object detection tasks. However, the sharply varying object scales, especially in RSI, can lead to inconsistency of IoU changes, which results in slow regression convergence and hinders the gradient propagation progress of bboxes. As shown in Fig. 2, the object scales in DIOR vary widely, significantly impacting both the regression procedure and detection accuracy. Thoughtfully, we statistic the object scale variation on DOTA and analyze the inherent relationship between object size and IoU values, speeding up the regression by adjusting a faster direction for gradient descent.

Fig. 5 intuitively depicts the diverse impacts that the same amount of bbox offset has on large/small-scale samples. Given a large bbox b_1 and a small bbox $b2$ which are proportional with the same IoU, the vertical side lengths l_1 and l_2 follow $l_1 < l_2$. When a slight vertical offset Δ occurs, the reduction ratios follow $r_1 = (\Delta/l_1)$ and $r_2 = (\Delta/l_2)$. It is obvious that $r_1 > r_2$, and the larger $l_2 - l_1$ is, the greater the variance in the magnitude of IoU changes, resulting in more pronounced gradient fluctuations. This example demonstrates that the overlap rate is sensitive to the bbox scales, and it can slump and drastically change gradient values, altering the direction of gradient descent. For samples with different IoU values that indicate the overlap rates, it is evident that bboxes with different scales bring about utterly different impacts. Two conclusions can be drawn.

1) For samples with high IoU, smaller adjunct bboxes possess a greater gradient change trend than actual anchors.
2) For samples with low IoU, larger adjunct bboxes possess a greater gradient change trend than actual anchors.

Consequently, SAL is introduced to leverage the strongly scale-related loss paradigm, compensating for the defects in existing losses. As depicted in Fig. 5, the GT b^{gt} and the adjunct GT $\hat{b}^{\mathrm{gt}} = [(\hat{x}_l^{\mathrm{gt}}, \hat{y}_t^{\mathrm{gt}}), (\hat{x}_r^{\mathrm{gt}}, \hat{y}_b^{\mathrm{gt}})]$ share the same bbox

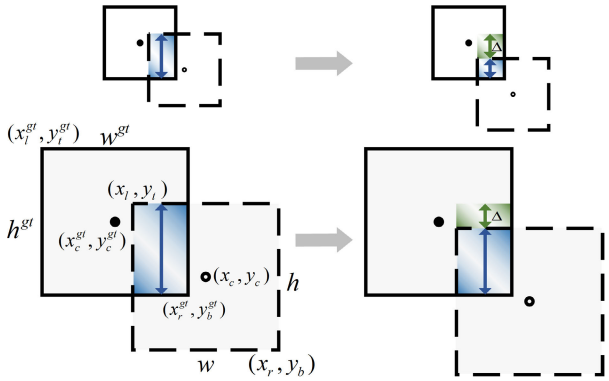


Fig. 5. Illustration of the influence of bbox offset on different IoU samples. The solid line rectangles and dashed line rectangles present the GT and anchor, respectively. The blue rectangles refer to the overlaps, i.e., IoU, between GT and anchor, and the green rectangles refer to the changed regions due to Δ.

center $(x_c^{\mathrm{gt}}, y_c^{\mathrm{gt}})$, and the anchor b and the adjunct anchor $\hat{b} = [(\hat{x}_l, \hat{y}_t), (\hat{x}_r, \hat{y}_b)]$ also share the same bbox center (x_c, y_c), where the subscripts l, r, t, b refer to left, right, top, and bottom, respectively. w^{gt} and h^{gt} are used to represent the width and height of GT, and w and h are for that of the anchor. In addition, we introduce an adaptive scaling ratio η to automatically adjust the transformation depending on the current IoU value, enhancing the model generalization

$$\hat{x}_l^{\mathrm{gt}} = x_c^{\mathrm{gt}} - \eta\frac{w^{\mathrm{gt}}}{2}, \quad \hat{y}_t^{\mathrm{gt}} = y_c^{\mathrm{gt}} - \eta\frac{h^{\mathrm{gt}}}{2} \tag{16}$$

$$\hat{x}_r^{\mathrm{gt}} = x_c^{\mathrm{gt}} + \eta\frac{w^{\mathrm{gt}}}{2}, \quad \hat{y}_b^{\mathrm{gt}} = y_c^{\mathrm{gt}} + \eta\frac{h^{\mathrm{gt}}}{2} \tag{17}$$

$$\hat{x}_l = x_c - \eta\frac{w}{2}, \quad \hat{y}_t = y_c - \eta\frac{h}{2} \tag{18}$$

$$\hat{x}_r = x_c + \eta\frac{w}{2}, \quad \hat{y}_b = y_c - \eta\frac{h}{2}. \tag{19}$$

With the tailored $\hat{b}^{\mathrm{gt}}$ and $\hat{b}$, the $\mathrm{IoU}^{\mathrm{sa}}$ can be described as

$$\begin{aligned} \mathrm{inter}^{\mathrm{sa}} = &\left(\min\left(\hat{x}_r^{\mathrm{gt}}, \hat{x}_r\right) - \max\left(\hat{x}_l^{\mathrm{gt}}, \hat{x}_l\right)\right) \\ &* \left(\min\left(\hat{y}_b^{\mathrm{gt}}, \hat{y}_b\right) - \max\left(\hat{y}_t^{\mathrm{gt}}, \hat{y}_t\right)\right) \end{aligned} \tag{20}$$

$$\begin{aligned} \mathrm{union}^{\mathrm{sa}} &= \left(\hat{w}^{\mathrm{gt}} * \hat{h}^{\mathrm{gt}} + \hat{w} * \hat{h}\right) * \eta^2 - \mathrm{inter}^{\mathrm{sa}} \\ \mathrm{IoU}^{\mathrm{sa}} &= \frac{\mathrm{inter}^{\mathrm{sa}}}{\mathrm{union}^{\mathrm{sa}}}. \end{aligned} \tag{21}$$

Taking CIoU loss L_{CIoU} as an example, the optimized scale-adaptive loss $L_{\mathrm{CIoU}}^{\mathrm{sa}}$ is obtained by introducing the IoU-based η

$$L_{\mathrm{CIoU}}^{\mathrm{sa}} = L_{\mathrm{CIoU}} + \mathrm{IoU} - \mathrm{IoU}^{\mathrm{sa}}. \tag{22}$$

It is derivable from 20 that when η is less than 1, the size of the adjunct bbox will be smaller than the actual bbox, resulting in a decrease in IoU which implies an increase in the gradient value and accelerating the regression convergence. Instead of manual adjustment, η is adaptively determined by the IoU value of the current sample.

IV. Experiments

A. Datasets and Evaluation Metrics

1) Datasets: Taking into account the specific requirements for the number of datasets in SSOD, we adopted two typical

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TABLE I
INSTANCE COUNT OF EACH CATEGORY FOR TRAINING ON DIOR

No.	Category	1%	5%	10%	20%
1	Airplane	13	55	128	252
2	Airport	2	19	37	71
3	Baseball field	7	90	134	290
4	Basketball court	6	31	51	105
5	Bridge	6	26	53	107
6	Chimney	7	22	36	57
7	Dam	2	11	30	48
8	Expressway service area	6	36	64	109
9	Expressway toll station	1	16	33	69
10	Golf field	1	23	30	51
11	Ground track field	7	30	54	121
12	Harbor	15	45	123	235
13	Overpass	6	19	47	101
14	Ship	120	486	1,186	2183
15	Stadium	1	16	30	64
16	Storage tank	23	47	98	238
17	Tennis court	27	112	193	448
18	Train station	2	8	25	42
19	Vehicle	57	311	646	1261
20	Windmill	7	37	58	154
	Total	316	1,440	3,056	6,006

TABLE II
INSTANCE COUNT OF EACH CATEGORY FOR TRAINING ON DOTA

No.	Category	1%	5%	10%	20%
1	Plane	97	474	1,074	2164
2	Baseball diamond	9	17	40	62
3	Bridge	6	57	99	245
4	Ground track field	4	5	13	24
5	Small vehicle	515	2,213	3,778	7,327
6	Large vehicle	104	922	2,760	6041
7	Ship	247	3,258	5,133	9,650
8	Tennis court	33	183	366	700
9	Basketball court	9	35	59	117
10	Storage tank	67	274	653	1614
11	Soccer ball field	3	13	23	32
12	Roundabout	4	4	12	32
13	Harbor	74	364	715	1622
14	Swimming pool	33	152	275	439
15	Helicopter	61	97	125	301
	Total	1,266	8,068	15,125	30,370

remote sensing datasets, DIOR and DOTA, which can meet the experimental needs for multiple different categories and target sizes, to validate the effectiveness and advancement of our proposed method.

a) DIOR: As a public collection of RSI specially designed for RSOD, DIOR consists of 23 463 images with 192 472 instances, spanning 20 specific categories of variable size. With the same image size of 800 × 800, the widely ranging spatial resolution, which varies from 0.5 to 30 m, contributes to apparent interclass resemblance and intraclass diversity. To be specific, the training set, validation set, and test set contain 58 625 863, and 11 738 images, respectively. The details of the categories can be found in Table I.

b) DOTA: Distinct from DIOR, DOTA contains 2806 images comprising 18 828 annotated objects of 15 common categories, and the image size ranges sharply from 800 × 800 to 4000 × 4000, giving rise to substantial impacts on detection accuracy and training stability. The detailed information of the categories is provided in Table II. It is worth noting that the images in DOTA are cropped to a uniform size of 1024 × 1024 with DOTA devkit to avoid affecting the detection accuracy of small objects.

2) Evaluation Metrics: To evaluate the performance of detectors, we introduce mAP_{50} as the critical metric, which sets the IoU threshold at 0.5 to filter unreliable predictions. $mAP_{50:95}$ is another evaluation metric, which represents the mAP across different thresholds ranging from 0.5 to 0.95 with an interval of 0.05, measuring the model's performance at multiple thresholds and presenting its robustness across varying IoU values. On account that $mAP_{50:95}$ is also commonly adopted for SSOD, we verified the models on DIOR and DOTA using both $mAP_{50:95}$ and mAP_{50}.

B. Detailed Implementation Setups

To ensure fairness and efficiency, we train all detectors on 1 NVIDIA GeForce RTX 3090 Ti GPU with five images (one labeled and four unlabeled), and SGD is employed as the optimizer with an initial learning rate of 0.002, momentum of 0.9, and weight decay rate of 0.0001. The unlabeled weight λ_u is set at 1.0 and the momentum of EMA is the same with [18] at 0.9995, and RetinaNet [39] is adopted as the teacher and student network. Following the generic data augmentation in consistent-teacher, we carry on using the default weak and strong augmentation for $\mathbf{D}_l$ and $\mathbf{D}_u$ separately, including random operations of resizing, flipping, translating, rotating, and erasing.

To fully evaluate the unlabeled knowledge exploration ability of SSOD methods, the datasets are redivided into combinations of a few labeled data and abundant unlabeled data. Specifically, we divide the DIOR and DOTA training sets into subsets under different protocols, 1%, 5%, 10%, and 20%, as labeled training data with annotations, and the rest serve as unlabeled training data without annotations. The validation sets of each dataset are used to test the methods. Additionally, the detectors are trained for 180 000 iterations under 1% protocol and 360 000 iterations under 5%, 10%, and 20% protocols to obtain well-trained models for reasonable experimental analysis. Particularly, the labeled data are sampled randomly under all protocols, which brings about the class imbalance problem, especially under the 1% protocol. To intuitively represent the variation of instance distribution induced by different protocols, Tables I and II count the instances of all categories in the training sets on DIOR and DOTA, respectively. Notably, when presenting and discussing the results on the specified dataset in the following context, we describe the categories with their number codes instead of full names. It is obvious that the number of instances in certain categories decreases significantly when using only 1% labeled data, leading to severe class imbalance and posing a tremendous challenge. The specific effects and analysis are elaborated in Section IV-C.

C. Experimental Results and Analysis

1) Comparison With the State-of-the-Art Methods: Since semi-supervised methods remain unexplored for HBB RSOD

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TABLE III
COMPARISON WITH OTHER METHODS ON DIOR AND DOTA

Dataset	Method	1%		5%		10%		20%	
		$mAP_{50:95}$	mAP_{50}	$mAP_{50:95}$	mAP_{50}	$mAP_{50:95}$	mAP_{50}	$mAP_{50:95}$	mAP_{50}
DIOR	supervised	5.5	14.6	16.7	35.6	24.0	44.4	30.3	52.3
	Unbiased-Teacher	18.0	29.8	42.4	60.8	48.3	68.6	50.9	71.8
	Consistent-Teacher	19.1	30.8	43.3	62.6	49.4	69.8	52.2	72.7
	Ours	**21.8**	**35.1**	**45.2**	**64.0**	**50.2**	**71.2**	**53.8**	**74.1**
DOTA	supervised	2.0	6.1	9.0	21.1	12.0	26.0	15.6	31.3
	Unbiased-Teacher	8.9	17.0	21.1	35.8	24.9	39.3	26.5	42.6
	Consistent-Teacher	10.2	18.4	22.6	36.6	26.0	40.8	28.6	43.9
	Ours	**12.0**	**22.0**	**23.1**	**37.4**	**26.8**	**41.9**	**30.0**	**44.5**

TABLE IV
ABLATION STUDIES OF MODULES IN UAT ON DIOR

EUS	MSF	SAL	1%		5%		10%		20%	
			$mAP_{50:95}$	mAP_{50}	$mAP_{50:95}$	mAP_{50}	$mAP_{50:95}$	mAP_{50}	$mAP_{50:95}$	mAP_{50}
			19.1	30.8	43.3	62.6	49.0	69.2	52.2	72.7
✓			20.8	33.7	44.6	63.6	49.0	69.3	52.7	73.2
	✓		21.8	35.0	43.8	62.9	49.1	70.0	52.8	73.4
		✓	20.2	32.5	43.5	63.1	49.5	69.4	52.6	73.3
✓	✓		21.7	34.4	44.1	63.4	49.3	69.5	53.1	73.6
✓		✓	21.0	35.0	44.2	63.3	49.8	70.2	53.5	73.7
	✓	✓	21.5	34.7	43.1	63.2	50.0	70.8	53.7	73.9
✓	✓	✓	**21.8**	**35.1**	**45.2**	**64.0**	**50.2**	**71.2**	**53.8**	**74.1**

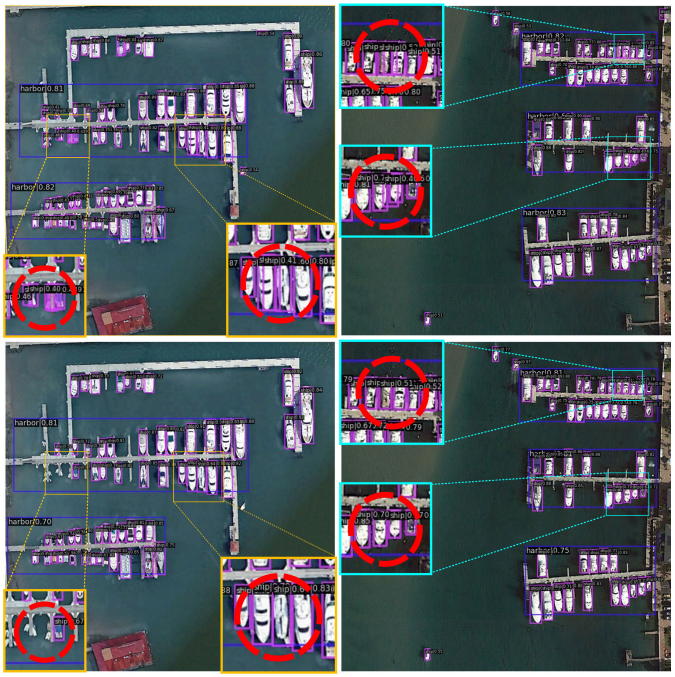


Fig. 6. Visualization of dense predictions in DIOR. The red circles mark the false detections caused by the feature confusion of consistent-teacher in the first row, and the results corrected by UAT are illustrated in the second row, proving the effectiveness of the proposed EUS.

and few of the existing works are open-sourced, we mainly compare our method with the state-of-the-art SSOD methods for natural images, including consistent-teacher [18], which is also set as our baseline, and unbiased-teacher [43]. In addition, we introduce RetinaNet as the supervised method to demonstrate the superior potential information extraction ability of SSOD methods. All methods are conducted on DIOR and DOTA datasets and evaluated by the $mAP_{50:95}$ and mAP_{50}.

In Table III, the experimental results demonstrate that UAT exhibits outstanding performance in contrast to the comparison algorithms, improving the detection accuracy under all protocol settings. On DIOR, UAT reaches 35.1%, 64.0%, 71.2%, and 74.1% at mAP_{50} with 1%, 5% 10%, and 20% labeled data, surpassing unbiased-teacher by and consistent-teacher by 4.3%, 1.4%, 1.4%, and 1.4%, respectively. UAT displays superior functionality on the metric $mAP_{50:95}$ as well, increasing the baseline by 2.7%, 1.9% 0.8%, and 1.6% on the whole. Compared with the supervised method, UAT achieves an increase over 20% on both mAP_{50} and $mAP_{50:95}$, revealing an evident detection ability.

Similarly, we conducted comparative experiments on DOTA to confirm the generalization ability of UAT. Table III demonstrates that UAT has a better performance with 22.0%, 37.4%, 41.9%, and 41.5% at mAP_{50} under all proportion setups, far surpassing the typical supervised method by 15.9%, 16.3%, 15.9%, and 13.2%, and outperforms the Unbiased-teacher and consistent-teacher by an average increase over 1.5%. Conclusively, our proposed method is proven effective in boosting detection performance.

2) Ablation Studies: In this part, we evaluate the impact of individual components in UAT on DIOR to prove their efficiency. For a fair comparison, we employ the same training/validation strategy and datasets that are utilized in comparison experiments. By removing specific modules in isolation, we can gain insights into their unique contributions. From the results in Table IV, we observe that our proposed model with all components achieves the highest mAP_{50} of 35.1%, 64.0%, 71.2%, and 74.1% at 1%, 5%, 10%, and 20% protocols, respectively. Removing EUS leads to an average

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TABLE V
PERFORMANCE OF INDIVIDUAL CATEGORY ON DIOR

Protocol	Method	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	mAP_{50}
1%	supervised	51.3	1.8	33.7	1.2	1.3	42.7	0.4	7.1	1.1	7.4	20.6	4.9	0.1	5.5	2.6	42.0	55.1	0.0	8.6	5.0	14.6
	Unbiased-Teacher	72.6	24.1	37.4	51.2	3.2	66.9	14.1	27.3	38.6	40.9	28.5	1.1	6.6	19.4	15.4	43.3	61.2	0.9	25.0	18.5	29.8
	Consistent-Teacher	74.7	25.3	**38.3**	52.1	3.4	**68.8**	15.9	28.9	**39.5**	41.1	29.9	1.3	7.3	19.8	15.7	44.0	62.5	1.4	26.4	19.8	30.8
	Ours	**79.5**	**28.1**	37.5	**55.4**	**7.1**	68.6	**19.6**	**35.0**	39.3	**45.9**	**33.8**	**10.7**	**13.8**	**28.4**	**19.4**	**47.3**	**69.2**	**7.6**	**30.8**	**24.7**	**35.1**
5%	supervised	79.9	16.2	85.4	44.4	2.5	79.3	9.5	35.9	20.3	45.0	31.2	8.2	8.5	26.7	24.9	67.2	75.1	0.8	31.9	19.8	35.6
	Unbiased-Teacher	89.6	61	90.2	75.4	17.9	86.9	40.3	63.4	50.0	69.8	49.6	31.9	43	70.6	79.2	68.8	82.4	19.5	64.5	62.1	60.8
	Consistent-Teacher	91.0	**63.5**	**91.1**	76.6	20.0	88.0	42.1	64.9	51.7	72.0	51.0	32.8	**44.2**	**73.7**	79.9	72.8	84.7	20.9	61.2	63.0	62.6
	Ours	**92.5**	60.2	90.9	**79.9**	**25.9**	**90.0**	**43.9**	**66.2**	**53.4**	**73.1**	**55.4**	**33.4**	42.6	72.3	**82.2**	**74.2**	**85.8**	**24.1**	**68.7**	**64.4**	**64.0**
10%	supervised	89.5	23.3	89.0	58.4	6.9	84.3	14.7	44.7	26.8	57.7	45.1	12.4	21.8	38.7	35.3	71.1	80.0	13.2	41.3	33.5	44.4
	Unbiased-Teacher	91.0	74.2	90.4	77.2	35.2	88.9	46.1	64.5	55.0	76.4	66.8	42.3	50.4	75.7	87.6	74.1	83.7	52.4	70.5	69.3	68.6
	Consistent-Teacher	91.8	75.3	**91.2**	78.8	36.5	89.4	47.0	65.8	**55.3**	77.2	68.0	43.2	53.9	77.6	89.0	75.4	84.9	**53.0**	71.4	**70.5**	69.8
	Ours	**93.5**	**76.2**	91.0	**81.9**	**37.0**	**91.5**	**47.1**	**67.2**	55.2	**78.6**	**71.9**	**42.7**	**55.9**	**79.3**	**91.4**	**78.0**	**88.9**	52.5	**71.9**	70.3	**71.2**
20%	supervised	90.5	38.8	90.4	63.0	12.4	86.7	26.6	51.1	38.4	58.0	52.3	19.1	30.7	51.5	70.2	72.8	82.2	18.3	45.5	48.3	52.3
	Unbiased-Teacher	91.5	79.6	90.7	80.7	38.9	88.8	53.9	65.4	58.7	77.6	75.5	42.2	53.2	77.6	92.3	78.6	87.1	58.3	70.7	75.4	71.8
	Consistent-Teacher	91.9	80.7	91.7	82.0	39.6	89.5	54.6	66.1	59.1	78.7	76.7	43.5	54.4	78.4	93.0	79.6	88.0	59.4	71.8	76.1	72.7
	Ours	**94**	**81.8**	**91.8**	**83.6**	**41.7**	**90.6**	**55.9**	**67.2**	**60.8**	**79.1**	**77.4**	**44.9**	**55.3**	**80.5**	**94.6**	**80.4**	**91.8**	**60.1**	**72.9**	**76.6**	**74.1**

TABLE VI
PERFORMANCE OF INDIVIDUAL CATEGORY ON DOTA

Protocol	Method	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	mAP_{50}
1%	supervised	21.5	0.1	0.0	0.1	4.6	5.8	5.4	29.3	0.1	6.7	0.0	5.3	2.4	4.9	5.2	6.1
	Unbiased-Teacher	25.4	1.0	0.1	1.8	10.5	14.9	9.7	58.6	7.0	56.6	0.1	4.9	16.2	30.8	17.1	17.0
	Consistent-Teacher	27.0	2.2	0.1	**2.1**	19.3	18.6	10.9	60.6	**7.5**	57.4	0.1	5.8	16.0	**31.5**	17.5	18.4
	Ours	**37.1**	**13.8**	**0.4**	1.0	**22.1**	**22.1**	**16.2**	**63.9**	6.3	**62.2**	**0.2**	**6.2**	**20.3**	30.7	**26.9**	**22.0**
5%	supervised	51.1	2.4	0.0	1.0	15.6	36.8	36.6	64.8	15.4	37.5	6.6	1.0	16.9	20.5	9.5	21.1
	Unbiased-Teacher	66.0	16.4	3.1	2.6	36.9	61.7	52.4	70.2	29.4	70.6	6.8	8.6	48.5	40.9	22.2	35.8
	Consistent-Teacher	**68.5**	16.5	4.0	**3.3**	**38.7**	62.2	53.7	**73.3**	29.6	70.9	6.9	9.1	48.8	41	22.4	36.6
	Ours	66.3	**19.4**	**4.5**	2.8	37.1	**64.2**	**54.2**	70.7	**32.8**	**71.7**	**9.0**	**11.6**	**49.0**	**41.7**	**25.9**	**37.4**
10%	supervised	60.7	11.0	0.0	2.1	19.7	46.8	40.1	71.6	16.7	47.3	8.6	3.4	26.6	30.7	5.2	26.0
	Unbiased-Teacher	71.2	13.9	3.3	12.0	34.8	71.0	53.7	77.9	34.6	71.5	8.2	8.0	53.6	47.1	28.8	39.3
	Consistent-Teacher	73.1	15.4	4.0	13.7	**35.1**	72.6	54.9	79.1	**35.4**	73.5	9.4	**13.5**	54.0	48.4	**30.2**	40.8
	Ours	**74.8**	**15.7**	**6.0**	**18.2**	34.6	**72.7**	**55.8**	**79.1**	34.5	**75.1**	**17.1**	8.3	**56.7**	**50.2**	28.9	**41.9**
20%	supervised	67.0	7.3	0.8	4.7	23.6	53.3	45.6	73.7	21.3	56.3	6.1	11.1	37.7	38.1	19.9	31.1
	Unbiased-Teacher	76.5	17.4	4.0	17.3	41.2	75.8	55.7	80.4	39.6	76.3	10.7	9.9	58.1	46.3	30.2	42.6
	Consistent-Teacher	77.1	18.6	5.6	18.1	41.9	76.4	56.4	82.9	40.4	78.2	12.0	11.4	59.1	48.7	31.2	43.9
	Ours	**77.2**	**19.0**	**6.0**	**18.5**	**42.3**	**77.0**	**56.9**	**83.0**	**40.9**	**78.6**	**17.5**	**12.1**	**60.2**	**49.0**	**31.3**	**44.5**

drop of 0.5% at mAP_{50}, indicating the interclass uncertainty exploration ability of EUS. Without the MSF strategy, the model has a slight decrease of 0.1% mAP_{50} at 1% protocol, while sharply declines 0.7%, 1.0%, and 0.4% at 5%, 10%, and 20% protocols, suggesting that the threshold iterated by multi-Gaussian-based sample fitting is beneficial, filtering out higher quality pseudo-bboxes. Finally, modifying the loss function without SAL results in dramatic mAP_{50} decrease of 0.7%, 0.6%, 1.7%, and 0.5%, demonstrating that not only can SAL accelerate the regression in SSOD process, but also conductive to the detection performance.

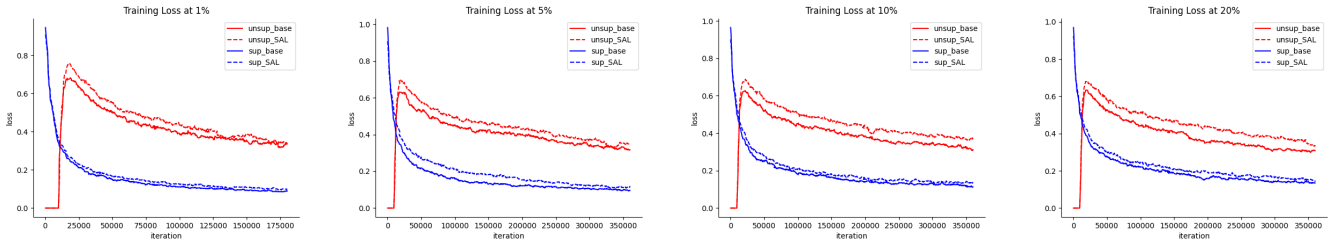
However, it is inevitable that the proposed modules have a certain degree of performance coupling as shown in Table IV, especially when there is an extreme lack of data. For example, the integration of EUS or SAL constrains the optimization of MSF under 1% protocol. Considering the design philosophies of different modules, this coupled phenomenon is mainly caused by inaccurate label softening and unexpected convergence disturbance, with the rather fixed and sensitive data distribution. Overall, although this phenomenon somehow constrains the modules to maximize their effectiveness, it is still worth pondering as we conduct further research in the future.

3) Analysis of Individual Category: In Tables V and VI, we demonstrate the superior performance of UAT with category-wise experimental results on DIOR and DOTA.

Overall, UAT performs well across all categories with varying degrees of improvement. However, at 1% protocol, the results of more than half of the categories fail to meet the average detection performance, which is significantly influenced by the severe class imbalance problem, as shown in Tables I and II. For example, as illustrated in Table VI, Category 3 (Bridge), Category 4 (Ground track field), Category 9 (Basketball court), Category 11 (Soccer ball field), and Category 12 (Roundabout) show relatively poor performance with only a few training instances. Moreover, categories with larger object scales consistently demonstrate rather poor performance across all protocols, highlighting the persistent challenge posed by multiscale variations. Consequently, the necessity to address both class imbalance and multiscale object adaptation becomes a critical consideration for our future research endeavors. Even though, UAT exhibits better performance depending on its reliable detection capability, confirming its effectiveness and superiority.

Comparing the results in Tables V and VI, it can be observed that at all protocols, UAT exhibits inconsistent

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(a) (b) (c) (d)

Fig. 7. Training bbox losses of baseline and UAT at (a) 1%, (b) 5%, (c) 10%, and (d) 20% protocols. In (a)–(d), red curves present the training loss of the unsupervised branch in the model, and the blue curves refer to that of the supervised branch. The solid lines represent baseline losses, while the dashed lines refer to losses modified by SAL. Each red curve shows a sharp increase because the unsupervised branch is not trained until the supervised branch has undergone a certain number of iterations.



Fig. 8. Examples of detection results on DIOR with our model trained under 10% protocol, including multiscale instances across most categories. The objects are located in scenarios encompassing cities, suburbs, airports, and water surfaces, fully demonstrating the scenario complexity in RSI.

performance trends on the categories that both datasets contain. For instance, for the baseball field/diamond category (which is Category 3 in DIOR while Category 2 in DOTA), UAT underperforms on DIOR but outperforms on DOTA compared with the baseline, which results from different data distribution and characteristics of datasets. Obviously, EUS learns distinct interclass relationships across datasets for different category compositions. Also, the divergent scenarios lead to distinct contextual information for instances of the same category, resulting in MSF to learn different intraclass relationships.

Notably, the dense arrangement severely influenced the performance of the model. For example, Category 7 (Ship) has the largest number of instances in DOTA, while its detection accuracy is not optimal under all protocols. Although this phenomenon is hard to avoid, UAT makes effective improvements in this regard. The detection accuracy of Category 7 gains +5.3%, +0.5%, +0.9%, and +0.5% mAP_{50} under 1%, 5%, 10%, and 20% protocols, respectively. As the first row shown in Fig. 6, it is common for RSOD models to confuse the densely arranged objects, which often manifests as higher false detection rates. With the optimization of UAT, the false detection boxes are eliminated, as illustrated in the second row.

4) Analysis of SAL: As mentioned in Section III, SAL adaptively alleviates the regression uncertainty. To establish the validity of SAL, we visualize the regression convergence process of baseline and UAT at all protocols. As shown in Fig. 7, both supervised and unsupervised losses are compared to confirm the efficacy of SAL. Obviously, the losses modified by SAL have a noticeable increase compared to the baseline, particularly for the unsupervised branches under all protocols. Meanwhile, the magnitude of the decline in loss curves increases to some extent, especially under 5% protocol, effectively demonstrating that SAL can accelerate the convergence speed of the model's regression task.

5) Visualization Results: To directly verify the effectiveness of UAT, Fig. 8 displays some representative visualization examples of the predictions on the DIOR dataset at 10% protocol. As we can see, UAT is capable of detecting multiscale objects and adapting to the scenario complexity.

Furthermore, the heatmaps of features output from the teacher's backbone of the baseline and UAT are visualized to confirm the model's legitimacy. As shown in Fig. 9, UAT focuses more on the interested objects and the surrounding context rather than the wide and complex background, improving the model's clustering ability compared with baseline.

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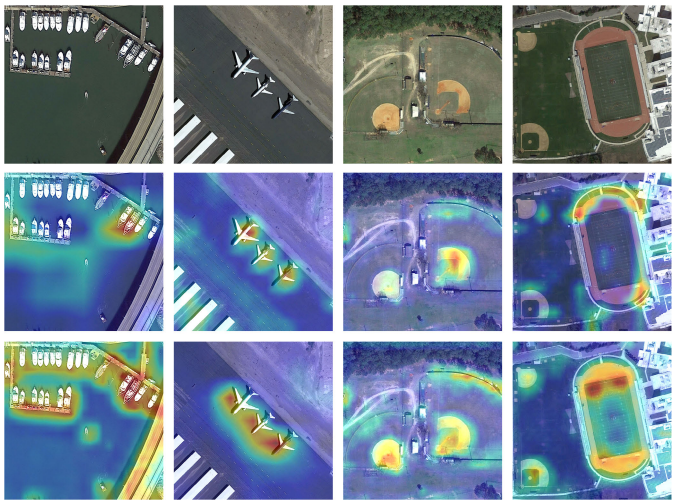


Fig. 9. Heatmaps of the inference results. The first row displays some original images in DIOR. The next two rows refer to the corresponding heatmaps of the features output from the backbones of baseline and UAT, respectively.

V. Conclusion

This article proposes an innovative TSM-based semi-supervised method, UAT, which excavates the potential of unlabeled data for efficient HBB RSOD. For classification uncertainty, EUS enhances interclass discrimination by identifying uncertain category pairs with chaotic clustering. MSF depicts the intraclass diversity by multi-Gaussian modeling of sub-category positive samples. For regression uncertainty, SAL optimizes the convergence consistency by tailoring scale-aware adjunct bboxes adaptively. Thus, combining these three modules, UAT constructs a novel paradigm for semi-supervised methods in RSOD, contributing to the further development of remote sensing.

References

[1] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 91–99.

[2] S. Zheng, Z. Wu, Y. Xu, and Z. Wei, "Instance-aware spatial-frequency feature fusion detector for oriented object detection in remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5606513.

[3] M. Cui, G. Ding, D. Yang, and Z. Chen, "DOPNet: Dense object prediction network for multiclass object counting and localization in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5604915.

[4] X. Qian, B. Wu, G. Cheng, X. Yao, W. Wang, and J. Han, "Building a bridge of bounding box regression between oriented and horizontal object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605209.

[5] G. Wang, H. Li, V. Sheng, Y. Ma, H. Ding, and H. Zhao, "DPMNet: A remote sensing forest fire real-time detection network driven by dual pathways and multidimensional interactions of features," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 1, pp. 783–799, Jan. 2025.

[6] J. Shen, C. Zhang, M. Zhang, Q. Li, and Q. Wang, "Learning remote sensing aleatoric uncertainty for semi-supervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5635413.

[7] X. Liang, J. Zhang, L. Zhuo, Y. Li, and Q. Tian, "Small object detection in unmanned aerial vehicle images using feature fusion and scaling-based single shot detector with spatial context analysis," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 6, pp. 1758–1770, Jun. 2020.

[8] X. Yang, S. Zhang, and W. Yang, "Two-way assistant: A knowledge distillation object detection method for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5612710.

[9] T. Zhang et al., "Multistage enhancement network for tiny object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611512.

[10] J. Li, P. Tian, R. Song, H. Xu, Y. Li, and Q. Du, "PCViT: A pyramid convolutional vision transformer detector for object detection in remote-sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5608115.

[11] Z. Ren, Y. Tang, Y. Yang, and W. Zhang, "SASOD: Saliency-aware ship object detection in high-resolution optical images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611115.

[12] C. Zhang, J. Su, Y. Ju, K.-M. Lam, and Q. Wang, "Efficient inductive vision transformer for oriented object detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5616320.

[13] J. E. van Engelen and H. H. Hoos, "A survey on semi-supervised learning," *Mach. Learn.*, vol. 109, no. 2, pp. 373–440, Feb. 2022.

[14] C. Chen, J. Song, C. Peng, G. Wang, and Y. Fang, "A novel video salient object detection method via semisupervised motion quality perception," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 5, pp. 2732–2745, May 2022.

[15] J. Wu, H. Fan, Z. Li, G.-H. Liu, and S. Lin, "Information transfer in semi-supervised semantic segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 2, pp. 1174–1185, Feb. 2024.

[16] Z. Li, Q. Lin, H. Fan, T. Zhao, and D. Zhang, "SIAVC: Semi-supervised framework for industrial accident video classification," 2024, *arXiv:2405.14506*.

[17] B. Chen et al., "Label matching semi-supervised object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14361–14370.

[18] X. Wang et al., "Consistent-teacher: Towards reducing inconsistent pseudo-targets in semi-supervised object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 3240–3249.

[19] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1195–1204.

[20] J. Shen, C. Zhang, Y. Yuan, and Q. Wang, "Enhancing prospective consistency for semi-supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2023, Art. no. 5619312

[21] X. Qian, C. Wang, W. Wang, X. Yao, and G. Cheng, "Complete and invariant instance classifier refinement for weakly supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5627713.

[22] T. Zhang et al., "Controllable generative knowledge-driven few-shot object detection from optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5612319.

[23] S. Gui, S. Song, R. Qin, and Y. Tang, "Remote sensing object detection in the deep learning era—A review," *Remote Sens.*, vol. 16, no. 2, p. 327, Jan. 2024.

[24] T. Zhang et al., "Posterior instance injection detector for arbitrary-oriented object detection from optical remote-sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5623918.

[25] T. Gao, Q. Niu, J. Zhang, T. Chen, S. Mei, and A. Jubair, "Global to local: A scale-aware network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5615614.

[26] C. Zhang, K.-M. Lam, and Q. Wang, "CoF-Net: A progressive coarse-to-fine framework for object detection in remote-sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600617.

[27] Y. Zhang, W. Guo, C. Wu, W. Li, and R. Tao, "FANet: An arbitrary direction remote sensing object detection network based on feature fusion and angle classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5608811.

[28] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

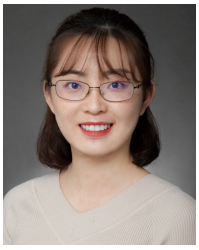
[29] T. Zhang, Y. Zhuang, G. Wang, H. Chen, L. Li, and J. Li, "A unified remote sensing object detector based on Fourier contour parametric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5611225.

[30] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Jan. 2023.

[31] K. Wang, F. Bai, J. Li, Y. Liu, and Y. Li, "MashFormer: A novel multiscale aware hybrid detector for remote sensing object detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 2753–2763, 2023.

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[32] Y. Tian, M. Zhang, J. Li, Y. Li, H. Yang, and W. Li, "FPNFormer: Rethink the method of processing the rotation-invariance and rotation-equivariance on arbitrary-oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5605610.
[33] H. Gao, S. Wu, Y. Wang, J. Y. Kim, and Y. Xu, "FSOD4RSI: Few-shot object detection for remote sensing images via features aggregation and scale attention," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 4784–4796, 2024.
[34] S. Antonelli et al., "Few-shot object detection: A survey," *ACM Comput. Surv.*, vol. 54, no. 11s, pp. 1–37, 2022.
[35] Q. Lu, T. Xu, J. Dong, Q. Liu, and X. Yang, "Improving object detection from remote sensing images via self-supervised adaptive fusion networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[36] T. Zhang et al., "Object-centric masked image modeling based self-supervised pretraining for remote sensing object detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1–13, 2023.
[37] X. Qian, Y. Huo, G. Cheng, C. Gao, X. Yao, and W. Wang, "Mining high-quality pseudoinstance soft labels for weakly supervised object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5607615.
[38] X. Xie, G. Cheng, X. Feng, X. Yao, X. Qian, and J. Han, "Attention erasing and instance sampling for weakly supervised object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5600910.
[39] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
[40] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren, "Distance-IoU loss: Faster and better learning for bounding box regression," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2020, vol. 34, no. 7, pp. 12993–13000.
[41] J. Wang and Y. Zhai, "Prototypical Siamese networks for few-shot learning," in *Proc. IEEE 10th Int. Conf. Electron. Inf. Emergency Commun. (ICEIEC)*, Jul. 2020, pp. 178–181.
[42] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.
[43] Y. Liu et al., "Unbiased teacher for semi-supervised object detection," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021, pp. 1–12.



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